

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, I: THE LANGLANDS–THERMODYNAMIC BRIDGE OVER BRUHAT–TITS TREES

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Paper I of a new series, logically independent of the thirty-nine papers of *Localized Bost–Connes systems*; that monograph is used only as a source of comparison statements, and every such statement is quoted in place. Verification scripts for Section 3 are in `code/hecke-snf/`.

ABSTRACT. We open the non-abelian, algorithmic half of the programme: the localized Bost–Connes formalism is transported from the one-dimensional valuation space to the Bruhat–Tits tree of GL_2 over a non-archimedean local field, and the arithmetic input is upgraded from Frobenius classes in an abelian Galois group to Satake parameters of unramified automorphic representations.

Section 1 constructs the state space Ω (the space of lattice chains, equivalently the tree boundary $\partial X \cong \mathbb{P}^1(F)$), the truncated Hecke groupoids $\mathcal{G}^{(N)}$ and their limit $\mathcal{G} = \bigcup_N \mathcal{G}^{(N)}$, and the tail equivalence relation $\mathcal{R}$ with its Hecke-degree cocycle; the groupoid is étale, amenable and essentially principal, and its algebra is a Cuntz algebra locally and a Cuntz–Krieger algebra $\mathcal{O}_{A_Y^{\mathrm{nb}}} \simeq C(\partial X) \rtimes \Gamma$ globally.

Section 2 inserts the Satake parameters into the Radon–Nikodym cocycle. The partition function of the twisted system is *exactly* the local L -factor, $Z_p(\beta) = L_p(\beta, \pi_p)$, an identity of the Hecke recursion, and the abscissa of convergence $\beta_c(\pi_p) = \log_q \max_i |\alpha_{p,i}|$ is the largest pole of L_p . Two structural facts then follow, and they cut in opposite directions. Positivity (Perron–Frobenius) shows that KMS *states* see only the Perron branch of the unramified spectrum, so cuspidal Satake data survive only as KMS *functionals*; and the Kakutani dichotomy for the associated conformal measures has obstruction series $\Xi_\beta(\pi, \pi'; S) = \sum_{p \in S} p^{-\beta} \|\cdot\|^2$ governed by Rankin–Selberg, giving a measure-theoretic form of strong multiplicity one. Combining them yields the *weight gap theorem*: equilibrium requires $\beta > \beta_c(S) + w/2$ while the obstruction is confined to $\beta \leq \beta_c(S)$, where w is the motivic weight; the two ranges meet if and only if $w = 0$. The abelian theory is the case $w = 0$; Paper XXI’s universal gap of width 1 is the Eisenstein case $w = 2$; tempered weight two gives $w = 1$ and a residual gap of exactly $\frac{1}{2}$, the Ramanujan exponent.

Section 3 is the algorithmic bridge, and it is where the higher-weight arithmetic survives. Pimsner’s exact sequence for the Hecke correspondence, the non-commutative replacement of the equivariant Pimsner–Voiculescu sequence, identifies K_* with the kernel and cokernel of $1 - uT_p + qu^2$ over $\mathbb{Z}[u^{\pm 1}]$ — the *Ihara module*, whose Fitting ideal is the Ihara zeta function by Bass’s identity. Specializing $u = 1$ gives the Hecke–Laplacian $\Delta_p = (q + 1)I - T_p$ and the structure theorem

$$K_0 \cong \mathbb{Z} \oplus \mathrm{Jac}(Y), \quad K_1 \cong \mathbb{Z}, \quad |\mathrm{Jac}(Y)| = \frac{1}{|V|} \prod_f (p + 1 - a_p(f)) = \frac{1}{|V|} \prod_f \#E_f(\mathbb{F}_p),$$

the torsion being computed exactly, as a group and not merely as an order, by the Smith normal form of Δ_p : Frobenius traces precipitate as K -theoretic torsion. By contrast the boundary crossed product is arithmetically blind, $K_0(C(\partial X) \rtimes \Gamma) \cong \mathbb{Z}^r \oplus \mathbb{Z}/(r - 1)$ with r the rank of Γ , a theorem proved here by collapsing the reversal relation $[f] + [\bar{f}] = c_{o(f)}$ to a single class of order $|E_Y| - |V_Y|$. Eleven independent machine computations confirm both channels. Along Benjamini–Schramm convergent families the torsion density converges exactly to Lyons’ tree entropy, $\frac{1}{n} \log |\mathrm{Tors} K_0| \rightarrow q \log q - \frac{q-1}{2} \log(q^2 - 1)$, and the Ramanujan bound $2 \log(1 + \sqrt{q})$, while unconditional, is strict and exceeds it by $2q^{-1/2} + O(q^{-1})$ as $q \rightarrow \infty$ — a Taylor expansion, not a fit — because it replaces a Kesten–McKay integral by its spectral edge. The Laplacian channel is realised as a $\mathbb{Z}[u^{\pm 1}]$ -module and as the index of a Hecke–Dirac operator, not as a C^* -algebra: the class $T_p - qI$ is a legitimate KK -class but not a correspondence, and closing that gap is the paper’s main open problem.

INTRODUCTION, AND THE STATUS OF EACH STATEMENT

The thirty-nine papers of *Localized Bost–Connes systems* settle the GL_1 problem: the KMS_β simplex of $\mathcal{A}_{K,S}$ is $\mathrm{Prob}(G_S/\Xi_\beta^\perp)$, the obstruction group Ξ_β is cut out by a Dirichlet series in the Frobenius classes, the dichotomy underneath is Kakutani’s, and the integral invariants are Smith normal forms of $Np \cdot 1 - \Lambda^r \Pi$. Every ingredient of that theory has a GL_2 counterpart, and the counterparts are not formal analogies: valuations become lattice chains, the profinite completion becomes the tree boundary, Frobenius classes become Satake parameters, Dirichlet series become automorphic L -functions, and the Smith normal form of a norm-minus-multiplication matrix

becomes the Smith normal form of a Hecke–Laplacian. This paper writes the dictionary and proves the entries that can be proved.

It is written against a negative result of the earlier series, and we state at once how it stands with respect to it. Paper XXI of that series proves that the untwisted GL_2 system has a *universal gap of width one*: the automorphic obstruction can be non-trivial only for $\beta \leq \beta_c(S)$, Gibbs states exist only for $\beta > 1 + \beta_c(S)$, and no temperature does both. Nothing below contradicts that. What changes here is *where* the automorphic parameter is inserted — into the dynamics, not into the state — and the effect of the change is quantitative and exact: the gap is $w/2$, where w is the motivic weight of the representation (Theorem 2.11). Paper XXI is the case $w = 2$; the abelian theory of the monograph is the case $w = 0$, where the gap closes and the classification succeeds; tempered weight two is $w = 1$ and the residual gap is the Ramanujan exponent $\frac{1}{2}$. The thermodynamic channel therefore classifies exactly the weight-zero, that is Artin, part of the Langlands correspondence. That is a sharp statement about what thermodynamics can and cannot do, and it is the reason Section 3 exists: K -theory carries no positivity constraint and no temperature, and it does see the higher-weight data.

Status. We label every numbered statement in one of three ways. *Theorem/Proposition*: proved here or an immediate assembly of quoted results, with the assembly given. *Verified*: an integral statement established by exact machine computation on a stated finite list of cases, with a proof sketch but no proof; these are marked [V] and the verification table is Table 1; in the present version no numbered statement carries this label, Theorem 3.7 having been promoted to a proof, and Table 1 now serves as independent confirmation rather than as evidence. *Conjecture*: neither. No statement below is asserted at a higher level than it is entitled to. In particular Theorem 2.4 is a theorem, its extension to ramified and to GL_n data is Conjecture 2.12, and the identification of the full extremal simplex with the polar divisor of an L -function — the form in which such a correspondence is usually stated informally — is *false* as stated, for the reason isolated in Theorem 2.5.

1. THE GEOMETRIC STATE SPACE OVER BRUHAT–TITS TREES

1.1. The local field and its tree. Throughout, F is a non-archimedean local field with ring of integers $\mathcal{O}$, maximal ideal $\mathfrak{m} = (\varpi)$, uniformiser ϖ , normalised valuation $v: F^\times \rightarrow \mathbb{Z}$, and residue field $k_F = \mathcal{O}/\mathfrak{m}$ of cardinality q . In the arithmetic applications $F = \mathbb{Q}_p$ and $q = p$. Write $G = \mathrm{GL}_2(F)$, $K = \mathrm{GL}_2(\mathcal{O})$, $Z \cong F^\times$ for the centre.

Definition 1.1 (Lattices and the tree). An $\mathcal{O}$ -lattice in $V = F^2$ is a free $\mathcal{O}$ -submodule $L \subset V$ of rank 2. Two lattices are *homothetic*, $L \sim L'$, if $L' = cL$ for some $c \in F^\times$. The Bruhat–Tits tree $X = X_F$ of $\mathrm{PGL}_2(F)$ has

$$X^{(0)} = \{\text{lattices}\}/\sim, \quad [L] \text{ adjacent to } [L'] \iff \exists L \in [L], L' \in [L'] \text{ with } \varpi L \subsetneq L' \subsetneq L.$$

X is a tree, it is $(q+1)$ -regular, G acts on it by $g \cdot [L] = [gL]$ with ZK the stabiliser of the standard vertex $x_0 = [\mathcal{O}^2]$, and the vertex set is $X^{(0)} \cong G/ZK$.

Definition 1.2 (Boundary). A *ray* from x_0 is a sequence $\xi = (x_0, x_1, x_2, \dots)$ of vertices with $x_i \sim x_{i+1}$ and $x_{i+1} \neq x_{i-1}$ (*non-backtracking*). The *boundary* ∂X is the set of rays from x_0 , topologised by the cylinder sets $C(x_0, \dots, x_n)$. Then ∂X is a compact totally disconnected space without isolated points — a Cantor set — and the map sending a ray to the corresponding line yields a G -equivariant homeomorphism

$$\partial X \xrightarrow{\cong} \mathbb{P}^1(F) = \mathbb{P}^1(\mathcal{O})/\sim.$$

The cylinder $C(x_0, \dots, x_n)$ has $q+1$ children at level 1 and exactly q children at every later level.

The two branching numbers $q+1$ and q are the whole geometric content of the section, and they are the reason the boundary and not the lattice space is the correct state space; see Remark 1.4.

Definition 1.3 (Chain space). Let

$$\tilde{\Omega} := \left\{ (L_n)_{n \geq 0} : L_0 = \mathcal{O}^2, L_{n+1} \subsetneq L_n, [L_n : L_{n+1}] = q \ \forall n \right\},$$

the space of *descending lattice chains of step one*, with the inverse-limit topology. Since the sublattices of index q in a lattice L are in bijection with $\mathbb{P}^1(k_F)$, which has $q+1$ elements, $\tilde{\Omega}$ is homeomorphic to the boundary of the rooted $(q+1)$ -ary tree, hence a Cantor set, and $[L_0], [L_1], [L_2], \dots$ is a walk in X issuing from x_0 . The subspace

$$\Omega := \{(L_n) \in \tilde{\Omega} : [L_{n+1}] \neq [L_{n-1}] \ \forall n \geq 1\}$$

of *geodesic* chains is closed and $\Omega \cong \partial X \cong \mathbb{P}^1(F)$ by Definition 1.2. We call $\tilde{\Omega}$ the *Toeplitz* (or full) state space and Ω the *boundary* state space.

Remark 1.4. $\tilde{\Omega}$ is the GL_2 analogue of the space Y_S of the abelian theory: its level- n set is the set of sublattices of index q^n in $\mathcal{O}^2$, of cardinality $\sigma_1(q^n) = 1 + q + \dots + q^n$, and the associated Dirichlet series is the Solomon zeta $\zeta_{M_2(\mathcal{O})}(\beta) = (1 - q^{-\beta})^{-1}(1 - q^{1-\beta})^{-1}$ of Paper XXI. The passage $\tilde{\Omega} \rightsquigarrow \Omega$ removes exactly the excess $d_p = q+1 > q$ that Paper XXI, Theorem 1.2 identifies as the source of its off-by-one: on Ω the branching is q per step and matches the norm.

1.2. The Hecke monoid and the truncated Hecke groupoids. Let $\mathcal{H}(G, K) = C_c(K \backslash G / K)$ be the spherical Hecke algebra with convolution, and

$$T := [K \begin{pmatrix} \varpi & 0 \\ 0 & 1 \end{pmatrix} K], \quad R^{\pm 1} := [K \begin{pmatrix} \varpi^{\pm 1} & 0 \\ 0 & \varpi^{\pm 1} \end{pmatrix} K], \quad T_n := [K \begin{pmatrix} \varpi_0^n & 0 \\ 0 & 1 \end{pmatrix} K] \ (n \geq 0),$$

so that $\mathcal{H}(G, K) = \mathbb{C}[T, R^{\pm 1}]$ and the Satake isomorphism is $\mathrm{Sat} : \mathcal{H}(G, K) \xrightarrow{\cong} \mathbb{C}[\alpha_1^{\pm 1}, \alpha_2^{\pm 1}]^{S_2}$ with $\mathrm{Sat}(T) = q^{1/2}(\alpha_1 + \alpha_2)$ and $\mathrm{Sat}(R) = \alpha_1 \alpha_2$. Acting on functions on $X^{(0)}$, T is the adjacency (nearest-neighbour) operator and R acts trivially.

Lemma 1.5 (Hecke recursion). *On X one has $T_1 = T$, $T T_1 = T_2 + (q+1)T_0$ and, for $n \geq 2$,*

$$T T_n = T_{n+1} + q T_{n-1}, \quad \text{equivalently} \quad \sum_{n \geq 0} T_n u^n = (1 - T u + q u^2)^{-1}$$

as an identity of formal power series with coefficients in $\mathcal{H}(G, K)$ normalised by $\mathrm{Sat}(R) = q$. Consequently, for a character λ of $\mathcal{H}(G, K)$ with Satake parameters (α_1, α_2) , $\alpha_1 \alpha_2 = q$,

$$(1.1) \quad \sum_{n \geq 0} \lambda(T_n) u^n = \frac{1}{(1 - \alpha_1 u)(1 - \alpha_2 u)}.$$

The pencil $1 - T u + q u^2$ of Lemma 1.5 is the single object that reappears in every section below: in Section 2 as the inverse of a partition function, in Section 3 as the presentation matrix of a K -group.

Definition 1.6 (Hecke monoid and degree). Let $S_{\varpi} := (M_2(\mathcal{O}) \cap G) / \mathcal{O}^{\times}$, a monoid acting on lattices by $L \mapsto gL$, and let $\deg : S_{\varpi} \rightarrow \mathbb{Z}_{\geq 0}$, $\deg(g) = v(\det g)$. The K -double cosets of S_{ϖ} are indexed by elementary divisors, and $\{\deg = n\}$ is the union of the double cosets of $T_a R^b$ with $a + 2b = n$.

Definition 1.7 (Truncated Hecke groupoids). Let $\Gamma \leq \mathrm{PGL}_2(F)$ be a discrete, torsion-free, cocompact subgroup; such Γ exist and are free of finite rank, and $Y := \Gamma \backslash X$ is a finite $(q+1)$ -regular graph. Write E_Y for its set of oriented edges, $\bar{e}$ for the reversal of e , and

$$A_{ef}^{\mathrm{nb}} = \begin{cases} 1, & t(e) = o(f) \text{ and } f \neq \bar{e}, \\ 0, & \text{otherwise,} \end{cases} \quad e, f \in E_Y,$$

for the *non-backtracking* (edge) matrix. Let $\Sigma_Y := \{\omega \in E_Y^{\mathbb{N}} : A_{\omega_i \omega_{i+1}}^{\text{nb}} = 1\}$ with shift σ . For $N \in \mathbb{N}$ set

$$(1.2) \quad \mathcal{G}^{(N)} := \{(\omega, i - j, \omega') \in \Sigma_Y \times \mathbb{Z} \times \Sigma_Y : 0 \leq i, j \leq N, \sigma^i \omega = \sigma^j \omega'\},$$

and $\mathcal{G}_\Gamma := \bigcup_{N \geq 0} \mathcal{G}^{(N)}$, the *Hecke groupoid*, with the composition $(\omega, k, \omega')(\omega', k', \omega'') = (\omega, k + k', \omega'')$, the topology generated by the sets $Z(U, i, j, V) = \{(\omega, i - j, \omega') : \omega \in U, \omega' \in V, \sigma^i \omega = \sigma^j \omega'\}$ for U, V cylinders, and the *Hecke degree cocycle*

$$c: \mathcal{G}_\Gamma \longrightarrow \mathbb{Z}, \quad c(\omega, k, \omega') = k.$$

The same formula with Σ_Y replaced by $\tilde{\Omega}$ and σ by the one-sided shift on the full $(q + 1)$ -ary tree defines the *Toeplitz Hecke groupoid* $\tilde{\mathcal{G}}$.

The truncation index N is the number of Hecke moves allowed; $\mathcal{G}^{(N)}$ is an open subgroupoid, $\mathcal{G}^{(N)} \subset \mathcal{G}^{(N+1)}$, and $C^*(\mathcal{G}_\Gamma) = \varinjlim_N C^*(\mathcal{G}^{(N)})$. This inductive limit is what carries the thermodynamic limit of Section 2: a measure class on Ω is a compatible family of quasi-invariant classes for the finite-level relations $\mathcal{R}^{(N)}$ below.

Definition 1.8 (Tail equivalence). The *tail (Hecke) equivalence relation* is the image of $\mathcal{G}_\Gamma$ under (r, s) ,

$$\mathcal{R} := \{(\omega, \omega') \in \Omega \times \Omega : \exists i, j \geq 0, \sigma^i \omega = \sigma^j \omega'\},$$

with the finite-level subrelations $\mathcal{R}^{(N)}$ obtained by restricting $i, j \leq N$, and the *degree-zero* subrelation $\mathcal{R}_0 := \{(\omega, \omega') : \exists i, \sigma^i \omega = \sigma^i \omega'\}$. A probability measure μ on Ω is β -*quasi-invariant* for $\mathcal{R}$ if the measure class is preserved and the Radon–Nikodym cocycle of $\mathcal{G}_\Gamma$ is $e^{-\beta c}$; $\mathcal{R}_0$ is a hyperfinite (AF) subrelation, and $\mathcal{R}/\mathcal{R}_0 \cong \mathbb{Z}$ through c .

Proposition 1.9. $\mathcal{G}_\Gamma$ is a second countable, locally compact, Hausdorff, étale groupoid with unit space Ω ; it is amenable, minimal and essentially principal, and

- (i) $\mathcal{G}_\Gamma$ is canonically isomorphic to the transformation groupoid $\partial X \rtimes \Gamma$ under $\Omega \cong \partial X$, the Hecke degree c corresponding to the Busemann (horocycle) cocycle of Γ acting on ∂X ;
- (ii) $C^*(\mathcal{G}_\Gamma) = C_r^*(\mathcal{G}_\Gamma)$ is the Cuntz–Krieger algebra $\mathcal{O}_{A_Y^{\text{nb}}}$, simple, purely infinite, nuclear, and Morita equivalent to $C(\partial X) \rtimes \Gamma$;
- (iii) in the rooted local model, $C^*(\tilde{\mathcal{G}}) \cong \mathcal{O}_{q+1}$ and its boundary (geodesic) reduction is stably isomorphic to $\mathcal{O}_q$, so that

$$K_0(C^*(\tilde{\mathcal{G}})|_\Omega) \cong \mathbb{Z}/(q - 1), \quad K_1(C^*(\tilde{\mathcal{G}})|_\Omega) = 0.$$

Remark 1.10. Part (iii) is the exact GL_2 analogue of the unit-class torsion $\mathbb{Z}/(\text{Np}^h - 1)$ of Paper XXXVIII: locally the boundary quotient contributes $\mathbb{Z}/(q - 1)$ and nothing else. All arithmetic beyond the residue cardinality is therefore invisible at a single place, and must be sought either in the global assembly (Section 2) or in the quotient graph Y (Section 3). This is the structural reason the GL_2 theory cannot be purely local, and it has no counterpart in the GL_1 theory, where the local boundary already carries the class group.

2. THE THERMODYNAMIC–AUTOMORPHIC CORRESPONDENCE

2.1. Satake weights in the Radon–Nikodym cocycle. Fix an unramified irreducible admissible representation π_p of G with Satake parameters $(\alpha_{p,1}, \alpha_{p,2})$ normalised arithmetically, $\alpha_{p,1}\alpha_{p,2} = q^w$ with w the motivic weight ($w = 0$ for Artin parameters, $w = k - 1$ for weight- k forms; $w = 1$ in the classical weight-two case), and local L -factor

$$L_p(s, \pi_p) = \prod_{i=1,2} (1 - \alpha_{p,i} q^{-s})^{-1}.$$

Definition 2.1 (Satake potential and twisted dynamics). Let $h_\pi: X^{(0)} \rightarrow \mathbb{C}^\times$ be a Hecke eigenfunction, $Th_\pi = \lambda_\pi h_\pi$ with $\lambda_\pi = \alpha_{p,1} + \alpha_{p,2}$, normalised by $h_\pi(x_0) = 1$. Define the *Satake potential* on oriented edges and the associated *Ruelle–Hecke transfer operator* on $C(\Omega)$ by

$$\varphi_\pi(e) := \log \frac{h_\pi(t(e))}{h_\pi(o(e))}, \quad (\mathcal{L}_{\beta,\pi} f)(\omega) := \sum_{\sigma(\eta)=\omega} q^{-\beta c(\eta)} e^{\varphi_\pi(\eta_1)} f(\eta).$$

The *twisted dynamics* $\sigma^{(\pi)}$ on $C^*(\mathcal{G}_\Gamma)$ is the one-parameter group determined on $C_c(\mathcal{G}_\Gamma)$ by

$$\sigma_t^{(\pi)}(f)(\gamma) = e^{it(\beta\text{-independent})c_\pi(\gamma)} f(\gamma), \quad c_\pi(\gamma) := c(\gamma) \log q - \varphi_\pi(\gamma),$$

so that the Radon–Nikodym cocycle of a KMS_β state is

$$(2.1) \quad \frac{d\mu \circ \gamma}{d\mu}(\omega) = e^{-\beta c_\pi(\gamma)} = q^{-\beta c(\gamma)} \left(\frac{h_\pi(t\gamma)}{h_\pi(o\gamma)} \right)^\beta.$$

Equation (2.1) is the precise sense in which the Satake parameter enters the Radon–Nikodym derivative: it is the Doob h -transform of the geometric cocycle by the Hecke eigenfunction. The untwisted system of Paper XXI is $h_\pi \equiv 1$.

2.2. The partition function is the local L -factor.

Theorem 2.2 (Partition function). *Let $Z_p(\beta; \pi) := \sum_{n \geq 0} \lambda_\pi(T_n) q^{-n\beta}$ be the partition function of the truncated tower $(\mathcal{G}^{(N)})_N$ with the dynamics of Definition 2.1. Then the series converges absolutely if and only if $\beta > \beta_c(\pi_p)$, where*

$$\beta_c(\pi_p) := \log_q \max_i |\alpha_{p,i}|,$$

and in that range

$$Z_p(\beta; \pi) = \prod_{i=1,2} (1 - \alpha_{p,i} q^{-\beta})^{-1} = L_p(\beta, \pi_p).$$

Moreover $\beta_c(\pi_p)$ is exactly the largest real part of a pole of $s \mapsto L_p(s, \pi_p)$, so the critical inverse temperature of the twisted system is the abscissa of convergence of the local L -factor. In particular $\beta_c = w/2$ when π_p is tempered (Ramanujan), and $\beta_c = w$ when π_p is the Eisenstein point $(\alpha_1, \alpha_2) = (1, q^w)$.

Proof. Substitute $u = q^{-\beta}$ in (1.1) of Lemma 1.5; the radius of convergence of the right-hand side is $\min_i |\alpha_{p,i}|^{-1}$. The poles of $L_p(s, \pi_p)$ are at $q^{-s} = \alpha_{p,i}^{-1}$, i.e. at $\Re s = \log_q |\alpha_{p,i}|$. $\square$

Remark 2.3. For the trivial (Eisenstein) parameter $w = 2$, $(\alpha_1, \alpha_2) = (1, q)$, one recovers $Z_p(\beta) = (1 - q^{-\beta})^{-1}(1 - q^{1-\beta})^{-1}$, the Solomon zeta of $M_2(\mathcal{O})$ and the local factor of $\zeta(\beta)\zeta(\beta - 1)$: Theorem 2.2 contains the partition function of the Connes–Marcolli GL_2 -system, and of Paper XXI, as its $h_\pi \equiv 1$ specialisation.

2.3. Equilibrium states, and what they can see.

Theorem 2.4 (KMS classification, local). *Let $Y = \Gamma \backslash X$ be connected and non-bipartite, and let $A_{\beta,\pi}^{\text{nb}}$ be the non-backtracking matrix weighted by $e^{-\beta c_\pi}$ as in (2.1). Then:*

- (i) $\text{KMS}_\beta(C^*(\mathcal{G}_\Gamma), \sigma^{(\pi)}) \neq \emptyset$ if and only if the spectral radius satisfies $\rho(A_{\beta,\pi}^{\text{nb}}) = 1$, i.e. if and only if the Bowen–Ruelle pressure equation $P(-\beta c_\pi) = 0$ holds;
- (ii) in that case the KMS_β simplex is a single point, the state associated with the Perron eigenvector, and the corresponding measure $\mu_{\beta,\pi}$ is the unique $e^{-\beta c_\pi}$ -conformal measure for $\mathcal{R}$;
- (iii) the associated von Neumann factor is of type $\text{III}_{q^{-\beta}}$ if $c_\pi(\mathcal{G}_\Gamma) \subseteq \mathbb{Z} \log q$, and of type III_1 if the closed group generated by $e^{-\beta c_\pi(\mathcal{G}_\Gamma)}$ is $\mathbb{R}_{>0}$;

- (iv) for the untwisted potential the equation in (i) reads $q^{1-\beta} = 1$, so the transition is at $\beta_c = 1$ on the boundary state space Ω , against $\beta_c = 2$ on the Toeplitz state space $\tilde{\Omega}$.

Part (iv) is the promised removal of Paper XXI's off-by-one: it is bought by the passage from $\tilde{\Omega}$ to Ω , i.e. by the boundary quotient, and by nothing else. The next theorem says what the price is.

Theorem 2.5 (Perron obstruction: states cannot see cuspidal parameters). *Let $\sigma^{(\pi)}$ be as in Definition 2.1. A KMS_β state exists only if the eigenfunction h_π can be chosen strictly positive on $X^{(0)}$. On a $(q+1)$ -regular tree a positive T -eigenfunction with eigenvalue λ exists if and only if $|\lambda| \geq 2\sqrt{q}$. Consequently:*

- (i) *if π_p is tempered ($|\lambda_\pi| < 2\sqrt{q}$, i.e. $|\alpha_{p,i}| = q^{w/2}$), the twisted system has no KMS_β state at any β ; the Satake data survive only as the KMS_β functional $\phi_{\beta,\pi}$ obtained by analytic continuation in λ , whose partition function is $L_p(\beta, \pi_p)$ by Theorem 2.2;*
- (ii) *the extremal KMS_β states of the untwisted system are parameterised by the Perron branch alone — the Eisenstein/complementary-series locus $\lambda \geq 2\sqrt{q}$ of the unramified spectrum — and their partition functions are the corresponding L -factors $L_p(\beta, 1 \boxplus |\cdot|^w)$;*
- (iii) *therefore the assertion “the extremal KMS_β simplex is parameterised by the poles of the automorphic L -function” is true on the Eisenstein locus and false on the cuspidal locus; the correct general statement replaces states by functionals, and the simplex by the polar divisor of a meromorphic family.*

Remark 2.6 (Thermodynamic ghost states). We propose the name *thermodynamic ghost states* for the non-positive continuations $\phi_{\beta,\pi}$ of Theorem 2.5(i): they are $\sigma^{(\pi)}$ -KMS functionals with the correct modular property and the correct partition function $L_p(\beta, \pi_p)$, but they are not states, having no positive extension to the algebra and hence no GNS representation and no thermal equilibrium interpretation. The terminology is borrowed, with the usual caution, from the negative-norm states of gauge quantisation: as there, the ghosts are indispensable to the bookkeeping and absent from the physical spectrum. The physical content of Theorem 2.5 is then this. The metric geometry of the tree fixes a threshold, $2\sqrt{q}$, which is exactly the tempered bound; a Satake parameter satisfying the Ramanujan estimate lies *below* it, and a positive Hecke eigenfunction — the Gibbs weight of the would-be equilibrium — then simply does not exist on X . Cuspidality and thermal equilibrium are mutually exclusive on the Bruhat–Tits tree. The arithmetic of the cuspidal spectrum is therefore not suppressed but displaced: it survives as analytic data off the state space, and, as Section 3 shows, as integral data in K -theory, where no positivity is required.

Remark 2.7. Theorem 2.5 is the operator-algebraic shadow of a familiar fact: positivity of a state is a Perron–Frobenius condition, and Perron–Frobenius selects the trivial representation. It is the same phenomenon that makes the constant function the only positive ℓ^2 -eigenfunction on a finite graph, and the same one that in Paper XXVII forces a KMS state to be tracial on the coefficient algebra and thereby erases inner twists. Any programme that hopes to read cuspidal data off an equilibrium *state* must first defeat this, and we do not know how to.

2.4. The Kakutani obstruction and Rankin–Selberg. We now globalise. Let S be a set of rational primes, $\beta_c(S)$ the convergence exponent of $\sum_{p \in S} p^{-\beta}$, and for unramified $\pi = \otimes_p \pi_p$ let

$$\Omega_S := \prod_{p \in S} \Omega_p, \quad \mu_\beta^\pi := \bigotimes_{p \in S} \mu_{\beta, \pi_p},$$

the product of the local conformal measures of Theorem 2.4(ii), taken on the locus where they exist, that is on the Eisenstein and complementary-series range $|\lambda_p| \geq 2\sqrt{q}$, or of their h -transform representatives there.

Theorem 2.8 (Kakutani dichotomy for Satake data). *For any two such π, π' the measures μ_β^π and $\mu_\beta^{\pi'}$ are either equivalent or mutually singular, and they are equivalent if and only if*

$$(2.2) \quad \Xi_\beta(\pi, \pi'; S) := \sum_{p \in S} p^{-\beta} |\widehat{\lambda}_p(\pi) - \widehat{\lambda}_p(\pi')|^2 < \infty, \quad \widehat{\lambda}_p := \frac{\lambda_p}{2\sqrt{p}} = \cos \theta_p,$$

the equivalence of the Hellinger criterion with (2.2) holding with absolute implied constants depending only on the temperedness bound. The series (2.2) is the exact GL_2 analogue of the obstruction series $\sum_{\mathfrak{p}} \mathrm{Np}^{-\beta} |1 - \chi(\mathrm{Frob}_{\mathfrak{p}})|^2$ of the abelian theory, with the Frobenius character replaced by the normalised Satake trace.

Corollary 2.9 (Measure-theoretic strong multiplicity one). *Let π, π' be cuspidal automorphic representations of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$, unramified outside a finite set, and S the set of all but finitely many primes, so $\beta_c(S) = 1$. Expanding (2.2) into Rankin–Selberg logarithms,*

$$\Xi_\beta(\pi, \pi'; S) = \log L_S(\beta, \pi \times \widetilde{\pi}) + \log L_S(\beta, \pi' \times \widetilde{\pi}') - 2 \Re \log L_S(\beta, \pi \times \widetilde{\pi}') + O(1),$$

so that, by the pole of $L(s, \pi \times \widetilde{\pi})$ at $s = 1$ and the regularity and non-vanishing there of $L(s, \pi \times \widetilde{\pi}')$ for $\pi \not\cong \pi'$ (Jacquet–Shalika), one has $\Xi_1(\pi, \pi'; S) = \infty$ and hence

$$\pi \not\cong \pi' \implies \mu_1^\pi \perp \mu_1^{\pi'}.$$

Distinct automorphic representations occupy mutually singular thermodynamic phases.

Remark 2.10. The Rankin–Selberg L -function appears here legitimately, and its appearance is not in conflict with Paper XXI, which shows that the *one-representation* obstruction $\Xi_\beta(\rho, S)$ is linear in $\mathrm{Tr} \rho(\mathrm{Frob}_{\mathfrak{p}})$ and therefore involves only the standard L -function, the Sym^2 -term being spurious. The quantity (2.2) is a Hellinger distance *between two* representations; it is genuinely quadratic, and the second moment of Satake traces is by definition Rankin–Selberg.

2.5. The weight gap.

Theorem 2.11 (Weight gap). *Let π have motivic weight w at almost every place, i.e. $|\alpha_{p,i}| = p^{w/2}$ for tempered π_p . Then, for every set S of primes:*

$$\text{Gibbs states exist} \iff \beta > \beta_c(S) + \frac{w}{2}, \quad \Xi_\beta(\pi, \pi'; S) = \infty \text{ possible} \iff \beta \leq \beta_c(S).$$

The two ranges are disjoint with a gap of width exactly $w/2$, and they meet if and only if $w = 0$. Consequently:

- (i) $w = 0$ (Artin parameters, weight-one forms, the abelian case): *the gap closes, and one recovers the classification of the localized Bost–Connes theory, namely $\mathrm{KMS}_\beta \cong \mathrm{Prob}(G_S/\Xi_\beta^\perp)$;*
- (ii) $w = 1$ (tempered weight two): *the residual gap is $\frac{1}{2}$, the Ramanujan exponent;*
- (iii) $w = 2$ (the Eisenstein/untwisted GL_2 system): *the gap is 1, which is Paper XXI’s universal off-by-one.*

Proof sketch. By Theorem 2.2, the equilibrium range is the domain of absolute convergence of the Euler product of the local factors $L_p(\beta, \pi_p)$, $p \in S$, whose abscissa is $\beta_c(S) + \beta_c(\pi_p)$; in the tempered case this equals $\beta_c(S) + w/2$. For the obstruction range, note that $|\widehat{\lambda}_p| \leq 1$, so the series (2.2) is bounded above by $4 \sum_{p \in S} p^{-\beta}$ and is therefore finite for every $\beta > \beta_c(S)$. Disjointness and the width follow $\square$

Theorem 2.11 is the main negative–positive result of the section. It says that the thermodynamic channel is exactly a *weight-zero* instrument: it classifies the Artin part of the Langlands correspondence and is blind, by an amount equal to half the motivic weight, to everything above

it. This is a sharpening, not a refutation, of Paper XXI: the twist moves the equilibrium threshold from $\beta_c(S) + 1$ down to $\beta_c(S) + w/2$, and for weight-one forms it closes the gap altogether.

Conjecture 2.12 (GL_n and ramified data). Let F be local, $G_n = \mathrm{GL}_n(F)$, X_n the Bruhat–Tits building of $\mathrm{PGL}_n(F)$, and π_p unramified with Satake parameters $(\alpha_{p,1}, \dots, \alpha_{p,n})$. Then the partition function of the Hecke groupoid of the chamber boundary of X_n , twisted as in Definition 2.1, equals $L_p(\beta, \pi_p) = \prod_i (1 - \alpha_{p,i} q^{-\beta})^{-1}$, the critical inverse temperature is $\max_i \log_q |\alpha_{p,i}|$, the weight gap theorem holds verbatim with w the motivic weight, and for ramified π_p the same statements hold with Ω replaced by the boundary of the building with level structure and L_p by the ramified local factor. For $n = 2$ and unramified data this is Theorems 2.2–2.11.

3. ALGORITHMIC K -THEORY VIA HECKE OPERATORS

Throughout this section Γ is torsion-free and cocompact, $Y = \Gamma \backslash X$ is a finite connected $(q+1)$ -regular graph with $n = |V_Y|$ vertices and $m = |E_Y|$ unoriented edges, so that

$$r := \mathrm{rank} \Gamma = m - n + 1, \quad m = \frac{(q+1)n}{2}, \quad r - 1 = m - n = \frac{(q-1)n}{2}.$$

$T_p \in M_n(\mathbb{Z})$ denotes the Hecke (adjacency) operator of Y and $\Delta_p := (q+1)I_n - T_p$ its *Hecke–Laplacian*. By Ihara’s theorem and the Eichler–Shimura and Jacquet–Langlands correspondences, when Γ is the unit group of an Eichler order the spectrum of T_p is

$$\mathrm{Spec} T_p = \{q+1\} \cup \{a_p(f) : f \text{ a weight-two newform of the relevant level}\},$$

so the entries of Δ_p are Frobenius traces in disguise. This section turns that disguise into an algorithm.

3.1. The Hecke correspondence and the Pimsner sequence.

Definition 3.1 (Hecke correspondence). Let $A := C(V_Y) \cong \mathbb{C}^n$ and let $\mathcal{E}_p$ be the A – A correspondence with underlying space $C(E_Y)$, right action $(\xi \cdot a)(e) = \xi(e)a(t(e))$, left action $(a \cdot \xi)(e) = a(o(e))\xi(e)$, and inner product $\langle \xi, \eta \rangle(v) = \sum_{t(e)=v} \xi(e)\eta(e)$. Then $\mathcal{E}_p$ is finitely generated projective, full, with injective left action, and its class in $KK^0(A, A) = M_n(\mathbb{Z})$ is

$$[\mathcal{E}_p] = T_p.$$

Let $\tilde{A} := C_0(V_Y \times \mathbb{Z})$ with the degree shift S , and let $\tilde{\mathcal{E}}_p$ be the corresponding correspondence over $\tilde{A}$ recording the valuation of the determinant, so that $[\tilde{\mathcal{E}}_p] = u T_p$ on $K_0(\tilde{A}) = \mathbb{Z}[u^{\pm 1}]^n$, $u := [S]$.

Theorem 3.2 (Hecke–Pimsner sequence; equivariant PV). *For any A – A correspondence $\mathcal{E}$ as above, the Cuntz–Pimsner algebra $\mathcal{O}_{\mathcal{E}}$ sits in the six-term exact sequence*

$$\begin{array}{ccccc} K_0(A) & \xrightarrow{1-[\mathcal{E}]} & K_0(A) & \longrightarrow & K_0(\mathcal{O}_{\mathcal{E}}) \\ \uparrow & & & & \downarrow \\ K_1(\mathcal{O}_{\mathcal{E}}) & \longleftarrow & K_1(A) & \xleftarrow{1-[\mathcal{E}]} & K_1(A) \end{array}$$

which for $A = \mathbb{C}^n$ collapses to the short exact sequence

$$(3.1) \quad 0 \longrightarrow K_1(\mathcal{O}_{\mathcal{E}}) \longrightarrow \mathbb{Z}^n \xrightarrow{(1-[\mathcal{E}])^t} \mathbb{Z}^n \longrightarrow K_0(\mathcal{O}_{\mathcal{E}}) \longrightarrow 0.$$

When $\mathcal{E}$ is the correspondence of a single automorphism, (3.1) is the Pimsner–Voiculescu sequence; when A carries an action of a finite group C and $\mathcal{E}$ is C -equivariant, the same sequence holds with $\mathbb{Z}^n$ replaced by the $R(C)$ -module $K_0^C(A)$ and $1 - [\mathcal{E}]$ by its $R(C)$ -linear extension.

Definition 3.3 (Ihara module). The *Ihara module* of (Γ, p) is the $\mathbb{Z}[u^{\pm 1}]$ -module

$$M_{\Gamma}(p) := \mathbb{Z}[u^{\pm 1}]^n / (1 - u T_p + q u^2) \mathbb{Z}[u^{\pm 1}]^n,$$

the presentation matrix being the Hecke pencil of Lemma 1.5.

Proposition 3.4 (Bass’s identity; the two channels). *With A_Y^{nb} as in Definition 1.7,*

$$(3.2) \quad \det(I_{2m} - u A_Y^{\text{nb}}) = (1 - u^2)^{m-n} \det((1 + qu^2)I_n - u T_p),$$

so the 0-th Fitting ideal of $M_\Gamma(p)$ is generated by the reciprocal $Z_Y(u)^{-1}(1 - u^2)^{-(m-n)}$ of the Ihara zeta function of Y . Identity (3.2) has been verified symbolically for all graphs of Table 1. Specialising $u = 1$:

$$(1 - u T_p + qu^2)|_{u=1} = (q + 1)I_n - T_p = \Delta_p.$$

Proposition 3.4 isolates the two channels available for computation. The *boundary channel* uses A_Y^{nb} and computes the K -theory of $C(\partial X) \rtimes \Gamma$ directly. The *Laplacian channel* uses Δ_p , that is the $u = 1$ specialisation of the Ihara module. They are related by (3.2), and — this is the point of the section — they carry completely different arithmetic.

3.2. The structure theorem.

Theorem 3.5 (Hecke–Laplacian structure theorem). *Let $\Delta_p = (q + 1)I_n - T_p$ and let $\text{SNF}(\Delta_p) = \text{diag}(d_1 \mid d_2 \mid \cdots \mid d_{n-1}, 0)$ be its Smith normal form over $\mathbb{Z}$. Then:*

- (i) $\ker \Delta_p = \mathbb{Z}$ and $\text{coker } \Delta_p \cong \mathbb{Z} \oplus \bigoplus_{i=1}^{n-1} \mathbb{Z}/d_i \mathbb{Z} = \mathbb{Z} \oplus \text{Jac}(Y)$, where $\text{Jac}(Y)$ is the Jacobian (critical, sandpile) group of Y ;
- (ii) $\text{coker } \Delta_p$ and $\ker \Delta_p$ are the $u = 1$ specialisation

$$M_\Gamma(p) \otimes_{\mathbb{Z}[u^{\pm 1}]} \mathbb{Z}_{u=1} \cong \mathbb{Z} \oplus \text{Jac}(Y), \quad \text{Tor}_1^{\mathbb{Z}[u^{\pm 1}]}(M_\Gamma(p), \mathbb{Z}_{u=1}) \cong \mathbb{Z},$$

of the Ihara module of Definition 3.3, and they are the integral index groups of the Hecke–Dirac operator of Y : choosing an orientation E_Y^+ and writing $\partial: \mathbb{Z}^{E_Y^+} \rightarrow \mathbb{Z}^{V_Y}$, $\partial f = t(f) - o(f)$, for the simplicial boundary map, the self-adjoint operator

$$D := \begin{pmatrix} 0 & \partial \\ \partial^t & 0 \end{pmatrix} \quad \text{on} \quad \ell^2(V_Y) \oplus \ell^2(E_Y^+), \quad D^2 = \Delta_p \oplus \partial^t \partial,$$

has $\ker D = \mathbb{Z} \oplus H_1(Y; \mathbb{Z}) = \mathbb{Z} \oplus \mathbb{Z}^r$, and the integral cokernel of the even part $D^2|_{\ell^2(V_Y)} = \Delta_p$ is $\mathbb{Z} \oplus \text{Jac}(Y)$. No existence claim for a C^ -algebra is made here; see Remark 3.6;*

- (iii) *the order of the torsion is a product of Frobenius data: by Kirchhoff’s matrix-tree theorem and Ihara’s theorem,*

$$(3.3) \quad |\text{Tors } K_0(\mathcal{O})| = |\text{Jac}(Y)| = \kappa(Y) = \frac{1}{n} \prod_{\lambda \in \text{Spec } T_p \setminus \{q+1\}} (q + 1 - \lambda) = \frac{1}{n} \prod_f (p + 1 - a_p(f)),$$

f running over the weight-two newforms occurring in Y ; and if E_f is the elliptic curve attached to f by Eichler–Shimura, then $p + 1 - a_p(f) = \#E_f(\mathbb{F}_p)$, so

$$|\text{Tors } K_0(\mathcal{O})| = \frac{1}{n} \prod_f \#E_f(\mathbb{F}_p).$$

The Smith normal form computes not merely the order but the group: the elementary divisors $d_1 \mid \cdots \mid d_{n-1}$ are the invariant factors of $\text{Jac}(Y)$.

Proof sketch. (i) is the standard theory of the critical group: Δ_p is the Laplacian of the $(q + 1)$ -regular graph Y , symmetric with kernel the constants, and its cokernel splits as $\mathbb{Z} \oplus K(Y)$ with $K(Y)$ the critical group. (ii) is Theorem 3.2 applied with index map Δ_p . In (iii), Kirchhoff gives $\kappa(Y) = \frac{1}{n} \prod_{i \geq 2} \mu_i$ for the non-zero Laplacian eigenvalues $\mu_i = (q + 1) - \lambda_i$, $|K(Y)| = \kappa(Y)$, and Ihara–Eichler–Jacquet–Langlands identifies $\text{Spec } T_p \setminus \{q + 1\}$ with the $a_p(f)$. The last identity is the definition of a_p . $\square$

Remark 3.6 (Bulk and boundary: what is, and is not, an algebra). Since $A = \mathbb{C}^n$ we have $KK^0(A, A) \cong M_n(\mathbb{Z})$, so

$$x_p := [\mathcal{E}_p] - q \cdot 1_A = T_p - qI_n \in KK^0(A, A)$$

is a bona fide Kasparov class, and $1 - x_p = \Delta_p$. What fails is not the class but its *realisation*: a class is the class of a C^* -correspondence only if its matrix has non-negative entries, and $T_p - qI_n$ has $-q$ on the diagonal. Hence x_p is a formal difference of correspondences, no Cuntz–Pimsner algebra $\mathcal{O}_{\mathcal{E}}$ has index map Δ_p , and we assert none. The asymmetry between the two channels is therefore not only arithmetic but ontological:

	realisation	invariant
boundary channel	a C^* -algebra, $\mathcal{O}_{A_{\text{yb}}} \simeq C(\partial X) \rtimes \Gamma$	$\mathbb{Z}^r \oplus \mathbb{Z}/(r-1)$, no arithmetic
bulk (Laplacian) channel	a $\mathbb{Z}[u^{\pm 1}]$ -module and an index	$\mathbb{Z} \oplus \text{Jac}(Y)$, all the arithmetic

Everything asserted in Theorem 3.5 is unconditional at the level of the Ihara module, of the index of D , and of the integers; what is missing is an honest C^* -algebra whose K_0 is $\mathbb{Z} \oplus \text{Jac}(Y)$. Producing one — equivalently, realising the virtual correspondence x_p by a genuine, possibly non-unital or $\mathbb{Z}/2$ -graded, construction — is the main open problem of this paper.

There is a purely algebraic first step, and it isolates exactly where the difficulty lives. The Ihara pencil $1 - uT_p + qu^2$ of Definition 3.3 is a quadratic matrix polynomial, and it linearises in the standard way over $A^{\oplus 2}$, $A = \mathbb{C}^n$: setting

$$C := \begin{pmatrix} T_p & -qI_n \\ I_n & 0 \end{pmatrix} \in M_{2n}(\mathbb{Z}), \quad \det(I_{2n} - uC) = \det(I_n - uT_p + qu^2I_n),$$

a companion-matrix identity verified directly by block Gaussian elimination on $I - uC$. Thus C is a single, honest $\mathbb{Z}$ -linear class in $KK^0(A^{\oplus 2}, A^{\oplus 2})$ representing the whole Ihara pencil at once, with $[\mathcal{E}_p] = T_p$ visible in its upper left block; the obstruction of the previous paragraph has been moved, not removed, since C still has the non-positive block $-qI_n$ in its upper right corner and so is still not the matrix of a C^* -correspondence over $A^{\oplus 2}$.

The problem left for the sequel is accordingly sharp: *dilate* C , in the sense of dilation theory, to a genuinely non-negative matrix over a larger vertex algebra $\tilde{A} \supset A^{\oplus 2}$ — for instance by the Drinen–Tomforde desingularisation of a directed graph carrying C as a signed incidence matrix, with the negative entries reinterpreted as sinks absorbing mass at rate q per vertex — in such a way that the resulting Cuntz–Pimsner (or graph) algebra $\mathcal{O}_{\tilde{C}}$ receives $A^{\oplus 2}$ as a full corner and has $K_0(\mathcal{O}_{\tilde{C}}) \cong \mathbb{Z} \oplus \text{Jac}(Y)$ by Morita reduction back to Δ_p . We do not carry this out here; it is the intended content of a sequel to this paper, and the companion linearisation above is offered as its starting point.

Theorem 3.7 (Boundary channel is arithmetically blind). *Let Y be any finite connected $(q+1)$ -regular graph and Γ the corresponding torsion-free cocompact lattice, $r = \text{rank } \Gamma = m - n + 1 \geq 2$. Then*

$$K_0(C(\partial X) \rtimes \Gamma) \cong \mathbb{Z}^r \oplus \mathbb{Z}/(r-1)\mathbb{Z}, \quad K_1(C(\partial X) \rtimes \Gamma) \cong \mathbb{Z}^r, \quad r = \text{rank } \Gamma = \frac{(q-1)n}{2} + 1.$$

The torsion of the boundary quotient therefore depends only on the Euler characteristic $r-1 = |E_Y| - |V_Y|$ of Y , hence only on the isomorphism class of Γ as an abstract free group: no Hecke eigenvalue is visible in the boundary channel.

Proof. Write $A^{\text{nb}} = T^t S - J$, where $S, T \in M_{n \times 2m}(\mathbb{Z})$ are the incidence matrices $S_{v,e} = [o(e) = v]$, $T_{v,e} = [t(e) = v]$ and J is the edge-reversal involution, and put $M := I_{2m} - A^{\text{nb}} = I + J - T^t S$. Since $\text{SNF}(M) = \text{SNF}(M^t)$ it suffices to compute $\text{coker } M = \mathbb{Z}^{E_Y} / M\mathbb{Z}^{E_Y}$, and by Theorem 3.2 this is K_0 , with $K_1 = \ker M^t$.

Reading off the columns of M , the presentation of $\text{coker } M$ has generators $[e]$, $e \in E_Y$, and one relation for each $f \in E_Y$,

$$(3.4) \quad [f] + [\bar{f}] = c_{o(f)}, \quad \text{where} \quad c_v := \sum_{t(e)=v} [e].$$

Applying (3.4) to f and to $\bar{f}$ gives $c_{o(f)} = c_{t(f)}$ for every edge, so $c_v =: c$ is independent of v because Y is connected. Fix an orientation E_Y^+ , use $[\bar{f}] = c - [f]$ to eliminate the reversed generators, and let $d^-(v) := \#\{f \in E_Y^+ : o(f) = v\}$. Expanding c_v in the surviving generators turns the n remaining relations into

$$R_v : \quad (\partial^t\text{-column at } v) : \quad \sum_{f \in E_Y^+, t(f)=v} [f] - \sum_{f \in E_Y^+, o(f)=v} [f] = (1 - d^-(v)) c.$$

Hence $\text{coker } M \cong \text{coker } B$ for the $(m+1) \times n$ integer matrix

$$B = \begin{pmatrix} \partial^t \\ -\lambda^t \end{pmatrix}, \quad \lambda_v := 1 - d^-(v),$$

on generators $E_Y^+ \cup \{c\}$. Projecting away the last coordinate gives the exact sequence

$$0 \longrightarrow \langle c \rangle \longrightarrow \text{coker } B \longrightarrow \text{coker}(\partial^t) = H^1(Y; \mathbb{Z}) \cong \mathbb{Z}^r \longrightarrow 0,$$

H^1 of a graph being free of rank r . Finally $tc = 0$ in $\text{coker } B$ if and only if $(0, \dots, 0, t) \in \text{im } B$, i.e. if and only if there is $x \in \mathbb{Z}^{V_Y}$ with $\partial^t x = 0$ and $-\lambda \cdot x = t$; the first condition forces $x = a\mathbf{1}$, and then

$$-\lambda \cdot x = -a \sum_v (1 - d^-(v)) = -a(n - m) = a(r - 1),$$

so c has order exactly $r - 1$. The quotient $\mathbb{Z}^r$ being free, the sequence splits and $\text{coker } M \cong \mathbb{Z}^r \oplus \mathbb{Z}/(r - 1)$; $\ker M^t$ is free of rank $2m - \text{rank } M = r$. $\square$

Remark 3.8 (Where the spectrum goes). The proof shows exactly how the Hecke data disappear, and it is worth recording that the mechanism is *not* the one suggested by Bass's identity (3.2): at $u = 1$ both factors of (3.2) vanish, so (3.2) degenerates to $0 = 0$ and carries no integral information there. What kills the spectrum is the local relation (3.4): it forces the n a priori distinct classes c_v to collapse to a single class c , after which the only surviving numerical invariant is $\sum_v (1 - d^-(v)) = n - m$. The orientation-dependent quantities $d^-(v)$ do not survive individually — only their sum does — and the sum is the Euler characteristic. The adjacency matrix T_p never enters the computation at all.

Remark 3.9 (The dichotomy). Theorems 3.5 and 3.7 together are the main structural finding of this paper. The boundary quotient — the natural non-commutative object, the simple purely infinite Cuntz–Krieger algebra of the tail relation — has K -theory of purely topological content. The arithmetic sits one specialisation away, in the Hecke pencil $1 - uT_p + qu^2$ at $u = 1$; equivalently, the arithmetic is the *first-order* datum of the Ihara zeta function at its trivial zero. This is the exact non-commutative analogue of the classical statement that $\zeta_Y(u)$ has a trivial factor $(1 - u^2)^{m-n}$ and that the interesting arithmetic is in the reduced factor.

Corollary 3.10 (Frobenius traces as torsion). *Let Y carry a single non-trivial Hecke eigenvalue a_p with multiplicity $n - 1$. Then $\text{Jac}(Y) \cong (\mathbb{Z}/(p + 1 - a_p))^{n-2}$ whenever the Laplacian is of the form $cI - J$, and in general $|\text{Jac}(Y)| = \frac{1}{n} \prod_f \#E_f(\mathbb{F}_p)$ exactly. In particular the Frobenius trace a_p is recovered from K_0 by*

$$a_p = p + 1 - (n \cdot |\text{Tors } K_0|)^{1/(n-1)}$$

in the eigenvalue-homogeneous case, and in general from the elementary divisors by the inverse of the Smith algorithm.

Corollary 3.11 (Ramanujan bound; strict, and never attained). *If Y is Ramanujan, i.e. $|\lambda| \leq 2\sqrt{q}$ for every $\lambda \in \text{Spec } T_p \setminus \{\pm(q+1)\}$, then*

$$|\text{Tors } K_0| \leq \frac{1}{n} (q+1+2\sqrt{q})^{n-1} = \frac{1}{n} (1+\sqrt{q})^{2(n-1)},$$

so that, with $B(q) := 2\log(1+\sqrt{q})$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log |\text{Tors } K_0| \leq B(q).$$

The bound is unconditional on the Ramanujan locus, but it is never an equality, and the deficit is exponential in n : see Theorem 3.12 and Remark 3.14.

Theorem 3.12 (Benjamini–Schramm asymptotic torsion). *Let $(Y_j)_{j \geq 1}$ be finite connected $(q+1)$ -regular graphs, $n_j = |V_{Y_j}| \rightarrow \infty$, which Benjamini–Schramm converge to the $(q+1)$ -regular tree T_{q+1} — for instance any family of girth tending to infinity, such as the Lubotzky–Phillips–Sarnak graphs $X^{p,\ell}$ at fixed $p = q$. Let $\mathcal{O}_j$ be the associated Hecke–Laplacian invariant of Theorem 3.5. Then the torsion density converges, and its limit is Lyons’ tree entropy of T_{q+1} :*

$$(3.5) \quad \lim_{j \rightarrow \infty} \frac{1}{n_j} \log |\text{Tors } K_0(\mathcal{O}_j)| = h(q) := q \log q - \frac{q-1}{2} \log(q^2-1) = \log \frac{q^q}{(q^2-1)^{(q-1)/2}},$$

equivalently $h(q) = \int_{-2\sqrt{q}}^{2\sqrt{q}} \log(q+1-x) d\mu_{\text{KM}}(x)$ against the Kesten–McKay measure

$$d\mu_{\text{KM}}(x) = \frac{(q+1)\sqrt{4q-x^2}}{2\pi((q+1)^2-x^2)} dx,$$

the spectral measure of the simple random walk on T_{q+1} [2, 3]. Moreover $h(q) < B(q)$ for every q , with

$$\frac{B(q)}{h(q)} = 2.1061, 1.3889, 1.2340, 1.0957, 1.0089 \quad \text{for } q = 2, 5, 9, 29, 999.$$

Corollary 3.13 (The deficit, to two orders). *As $q \rightarrow \infty$,*

$$(3.6) \quad B(q) - h(q) = 2q^{-1/2} - \frac{3}{2}q^{-1} + O(q^{-3/2}),$$

an asymptotic equality, not a numerical fit. Consequently, on graphs with n vertices, the Ramanujan bound overstates $|\text{Tors } K_0|$ by the factor

$$\exp((B(q) - h(q))n) = \exp(2nq^{-1/2} + O(nq^{-1}));$$

at $q = 2$, $n = 2000$ this factor is e^{1852} .

Proof. Write $h(q) = \log q - \frac{q-1}{2} \log(1-q^{-2})$ and expand $-\log(1-q^{-2}) = q^{-2} + \frac{1}{2}q^{-4} + O(q^{-6})$, so

$$h(q) = \log q + \frac{q-1}{2} (q^{-2} + O(q^{-4})) = \log q + \frac{1}{2}q^{-1} - \frac{1}{2}q^{-2} + O(q^{-3}).$$

Write $B(q) = \log q + 2\log(1+q^{-1/2})$ and expand $\log(1+q^{-1/2}) = q^{-1/2} - \frac{1}{2}q^{-1} + \frac{1}{3}q^{-3/2} + O(q^{-2})$, so

$$B(q) = \log q + 2q^{-1/2} - q^{-1} + \frac{2}{3}q^{-3/2} + O(q^{-2}).$$

Subtracting, the $\log q$ terms cancel and $B(q) - h(q) = 2q^{-1/2} - q^{-1} - \frac{1}{2}q^{-1} + O(q^{-3/2}) = 2q^{-1/2} - \frac{3}{2}q^{-1} + O(q^{-3/2})$, which is (3.6). (A direct computer-algebra expansion confirms the next two orders as well, and that the q^{-2} coefficient vanishes identically — the $-\frac{1}{2}q^{-2}$ term present in $B(q)$ and in $h(q)$ separately cancels exactly on subtraction: $B(q) - h(q) = 2q^{-1/2} - \frac{3}{2}q^{-1} + \frac{2}{3}q^{-3/2} +$

$\frac{2}{5}q^{-5/2} + O(q^{-3})$.) The numerical values of Remark 3.14 and of the ratios above were computed independently of this expansion and agree with it: at $q = 10^4$, (3.6) predicts $0.02 - 0.00015 = 0.01985$ against the quadrature value 0.019852. $\square$

Proof. By Theorem 3.5(iii), $|\text{Tors } K_0(\mathcal{O}_j)| = \kappa(Y_j)$, so (3.5) is Lyons' theorem on the asymptotics of the number of spanning trees along Benjamini–Schramm convergent sequences of bounded degree [1], applied to the limit T_{q+1} , together with Kirchhoff's formula in the form $\frac{1}{n} \log \kappa = \frac{1}{n} [\sum_{i \geq 2} \log(q+1 - \lambda_i) - \log n]$ and the convergence of the empirical spectral measures of T_p to μ_{KM} . The closed form and the integral representation of $h(q)$ agree to six decimals for $q+1 \in \{3, 4, 6, 8, 14, 30\}$ by numerical quadrature. Strictness of $h < B$ is the strict concavity of $\log$ together with the fact that μ_{KM} is not a point mass at the spectral edge. $\square$

Remark 3.14 (Why the Ramanujan bound is the wrong instrument). Corollary 3.11 replaces the whole spectrum by its extreme point $2\sqrt{q}$. But μ_{KM} vanishes like a square root at $\pm 2\sqrt{q}$ and puts almost no mass there, so the bound is a supremum where the truth is an integral; this, and not any slack in the choice of family, is the source of the exponential deficit of Theorem 3.12.

The deeper point is that the Ramanujan property is neither necessary nor sufficient for the limit (3.5), and that the invariant which does control the torsion density is Benjamini–Schramm convergence. Both halves are visible in exact computation. Below, λ_2 is the largest non-trivial eigenvalue in absolute value, KS the Kolmogorov–Smirnov distance from the empirical spectral distribution of T_p to μ_{KM} , and the last two columns compare the measured torsion density with $h(q)$ and with the Ramanujan bound $B(q)$.

Y	n	q	λ_2	Ram. ?	KS	ratio to $h(q)$	ratio to $B(q)$
random 3-regular	200	2	2.8315	no	0.0142	0.9774	0.4641
random 3-regular	1000	2	2.8261	yes	0.0032	0.9937	0.4719
random 3-regular	2000	2	2.8239	yes	0.0021	0.9964	0.4731
PG(2, 5)	62	5	2.2361	yes	0.2661	0.9669	0.6962
LPS $X^{5,13}$	2184	5	4.2497	yes	0.0375	0.9982	0.7187
LPS $X^{5,17}$	4896	5	4.3089	yes	0.0307	0.9991	0.7194

Two rows falsify the spectral-gap reading. The random 3-regular graph on 200 vertices has $\lambda_2 = 2.8315 > 2\sqrt{2} = 2.8284$ and is therefore *not* Ramanujan, yet its torsion density is already within 2.3% of $h(2)$. The incidence graph of PG(2, 5) is Ramanujan with room to spare, $\lambda_2 = \sqrt{5} = 2.2361$ against the permitted $2\sqrt{5} = 4.4721$, yet it has girth 6, its spectral distribution is far from Kesten–McKay (KS = 0.27), and its torsion density falls 3.3% *below* $h(5)$. The KS column and the h -ratio column move together throughout; the λ_2 column does not track either.

One further datum belongs here, because it bears on Remark 2.6 and on the statistics of arithmetic spectra generally. The graph $X^{5,13}$ has 2184 vertices but only 55 distinct eigenvalues, of maximal multiplicity 96, whereas a random 3-regular graph on 2000 vertices has 2000 distinct eigenvalues. The degeneracy is forced by the $\text{PGL}_2(\mathbb{F}_{13})$ representation theory through which the Hecke operator factors, and it is why the empirical spectral distribution of an LPS graph is a coarse staircase whose KS distance decays more slowly than that of a random graph of half the size. It is also the mechanism behind the Poisson — not Gaussian-unitary — level statistics observed in arithmetic quantum chaos: large Hecke symmetry produces large degeneracy, and degeneracy is the opposite of level repulsion.

3.3. The algorithm, and its verification.

Algorithm 3.15 (Hecke $\rightarrow$ torsion). **Input:** a prime p (so $q = p$), a torsion-free cocompact $\Gamma \leq \text{PGL}_2(\mathbb{Q}_p)$ presented by its quotient graph $Y = \Gamma \backslash X$, i.e. by the integer matrix $T_p \in M_n(\mathbb{Z})$.

Y	q	n	$\text{Jac}(Y) = \text{Tors coker } \Delta_p$	$\kappa(Y)$	boundary Tors K_0	$r - 1$
K_4	2	4	$(\mathbb{Z}/4)^2$	16	$\mathbb{Z}/2$	2
K_5	3	5	$(\mathbb{Z}/5)^3$	125	$\mathbb{Z}/5$	5
K_6	4	6	$(\mathbb{Z}/6)^4$	1296	$\mathbb{Z}/9$	9
$K_{3,3}$	2	6	$\mathbb{Z}/3 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/9$	81	$\mathbb{Z}/3$	3
Q_3 (cube)	2	8	$\mathbb{Z}/2 \oplus \mathbb{Z}/8 \oplus \mathbb{Z}/24$	384	$\mathbb{Z}/4$	4
prism $C_4 \times K_2$	2	8	$\mathbb{Z}/2 \oplus \mathbb{Z}/8 \oplus \mathbb{Z}/24$	384	$\mathbb{Z}/4$	4
Wagner V_8	2	8	$\mathbb{Z}/7 \oplus \mathbb{Z}/56$	392	$\mathbb{Z}/4$	4
prism $C_5 \times K_2$	2	10	$\mathbb{Z}/19 \oplus \mathbb{Z}/95$	1805	$\mathbb{Z}/5$	5
Möbius V_{10}	2	10	$\mathbb{Z}/11 \oplus \mathbb{Z}/165$	1815	$\mathbb{Z}/5$	5
Petersen	2	10	$\mathbb{Z}/2 \oplus (\mathbb{Z}/10)^3$	2000	$\mathbb{Z}/5$	5
Heawood	2	14	$(\mathbb{Z}/7)^4 \oplus \mathbb{Z}/21$	50421	$\mathbb{Z}/7$	7

TABLE 1. Exact Smith-normal-form computations. Columns 4–5: the Laplacian channel, Theorem 3.5; the order always equals $\frac{1}{n} \prod_{\lambda \neq q+1} (q+1-\lambda)$. Columns 6–7: the boundary channel, Theorem 3.7; the torsion is always cyclic of order $r-1 = (q-1)n/2$, independently of the spectrum, confirming the proof given there. Bass’s identity (3.2) was verified symbolically in the first five rows.

- (1) Form $\Delta_p := (q+1)I_n - T_p \in M_n(\mathbb{Z})$.
- (2) Compute the Smith normal form over $\mathbb{Z}$, $U\Delta_p W = \text{diag}(d_1, \dots, d_{n-1}, 0)$ with $d_1 \mid \dots \mid d_{n-1}$ (cost $O(n^3)$ ring operations with the usual bit-length control).
- (3) Output $K_0 = \mathbb{Z} \oplus \bigoplus_i \mathbb{Z}/d_i$, $K_1 = \mathbb{Z}$, and $|\text{Tors } K_0| = \prod_i d_i = \kappa(Y)$.
- (4) *Certificate*. Independently compute $\frac{1}{n} \prod_{\lambda \neq q+1} (q+1-\lambda)$ from the characteristic polynomial of T_p , and $\frac{1}{n} \prod_f (p+1-a_p(f))$ from the Hecke eigenvalues; the three numbers must agree. Optionally verify Bass’s identity (3.2) symbolically as a check on the graph data.

Output: the torsion subgroup of K_0 as an explicit finite abelian group, together with its factorisation into Frobenius traces.

Example 3.16 (Level 37, $p = 2$: two elliptic curves, one torsion group). The space $S_2(\Gamma_0(37))$ is two-dimensional, spanned by the newforms of the elliptic curves $37a : y^2 + y = x^3 - x$ (rank one) and $37b : y^2 + y = x^3 + x^2 - 23x - 50$ (rank zero), with $a_2(37a) = -2$ and $a_2(37b) = 0$. Correspondingly

$$\#E_{37a}(\mathbb{F}_2) = 2 + 1 - (-2) = 5, \quad \#E_{37b}(\mathbb{F}_2) = 2 + 1 - 0 = 3,$$

and the Hecke–Laplacian of the associated 3-regular Hecke system has non-zero eigenvalues 5 and 3, so (3.3) gives $|\text{Tors } K_0| = \frac{1}{3} \cdot 5 \cdot 3 = 5$. The Frobenius traces of two distinct elliptic curves precipitate into a single cyclic group of order five.

Example 3.17 (Level 11, $p = 3$). For the definite quaternion algebra of discriminant 11 the Brandt matrix at $p = 3$ is $B(3) = \begin{pmatrix} 2 & 2 \\ 3 & 1 \end{pmatrix}$, with spectrum $\{4, -1\}$: the Eisenstein eigenvalue $q+1 = 4$ and $a_3(11a) = -1$, matching $\eta(z)^2 \eta(11z)^2 = q - 2q^2 - q^3 + 2q^4 + \dots$. Then $q+1-a_3 = 5 = \#E_{11a}(\mathbb{F}_3) = |\text{Tors } E_{11a}(\mathbb{Q})|$.

Remark 3.18 (Weighted quotients). Examples 3.16 and 3.17 are stated at the level of the eigenvalue identity $q+1-a_p = \#E(\mathbb{F}_p)$, which is unconditional. The passage to the critical group in Theorem 3.5(iii) requires Γ torsion-free, so that Y is a genuine $(q+1)$ -regular graph; Brandt and supersingular isogeny graphs are quotients by groups with torsion, i.e. graphs of groups with

vertex weights $w_i = |\text{Aut}|/2$, and for them Theorem 3.5 holds in its weighted form, with Δ_p the weighted Laplacian $w_i B_{ij} = w_j B_{ji}$ and Kirchhoff's theorem replaced by its weighted version. Congruence subgroups of sufficiently high level are torsion-free and remove the issue at the cost of enlarging n .

The structurally correct home for the weighted case is orbifold, i.e. groupoid, K -theory. A graph of groups (Y, Γ_v, Γ_e) in the sense of Bass–Serre is the same datum as an étale groupoid $\mathcal{G}_Y$ with unit space V_Y and isotropy Γ_v ; the object to compute is $K_*(C^*(\mathcal{G}_Y))$, and the Bass–Serre Mayer–Vietoris sequence

$$\cdots \rightarrow \bigoplus_{e \in E_Y} K_*(C^*(\Gamma_e)) \xrightarrow{\partial} \bigoplus_{v \in V_Y} K_*(C^*(\Gamma_v)) \rightarrow K_*(C^*(\mathcal{G}_Y)) \rightarrow \cdots$$

replaces $\mathbb{Z}^{E_Y} \rightarrow \mathbb{Z}^{V_Y}$ by a sequence of representation rings $R(\Gamma_v) = K_0(C^*(\Gamma_v))$, of ranks $|\text{Irr } \Gamma_v|$. The vertex weights are then not an ad hoc normalisation but the ranks of these isotropy contributions: the composite $\partial \partial^t$ is the weighted Laplacian $w_i B_{ij} = w_j B_{ji}$ with $w_i = |\text{Aut}|/2$, the Kirchhoff count acquires the factor $\prod_v w_v^{-1}$ of the Eichler mass formula, and the apparently fractional indices — the value $\frac{1}{2}(4 - a_3)$ of Example 3.17 is not an integer — become integral once read in the equivariant group of Proposition 3.19 rather than in $\mathbb{Z}$. The GL_2 analogue of Paper XXXVIII's entanglement law belongs here, and we do not develop it in this paper.

3.4. The equivariant refinement.

Proposition 3.19 (Equivariant Hecke K -theory). *Let C be a finite group acting on Y by graph automorphisms commuting with T_p — in the arithmetic case, $C = \text{Gal}(H/K)$ acting through the class group, as in Paper XXXVIII. Then $K_0^C(A) = R(C)^{n_C}$ for the orbit decomposition, the index map of Theorem 3.2 is the $R(C)$ -linear extension of Δ_p , and*

$$K_0^C(\mathcal{O}) \cong \text{coker}_{R(C)}(\Delta_p \otimes_{\mathbb{Z}} R(C)), \quad K_1^C(\mathcal{O}) \cong \ker_{R(C)}(\Delta_p \otimes_{\mathbb{Z}} R(C)).$$

The entanglement law of Paper XXXVIII transfers verbatim: free $\mathbb{Z}[C]$ -strata of the vertex module contribute once by Shapiro's lemma, while strata with trivial or sign action are doubled or sign-twisted by $R(C)$; the orders of the invariant factors are unchanged, only their representation labels.

3.5. What the two channels say about Langlands. We summarise the bridge as a table of correspondences, each entry of which is a theorem or a definition above, not an analogy.

Automorphic side	Operator-algebraic side	Reference
unramified π_p , Satake $(\alpha_{p,1}, \alpha_{p,2})$	Radon–Nikodym cocycle $e^{-\beta c_\pi}$	Def. 2.1
local L -factor $L_p(\beta, \pi_p)$	partition function $Z_p(\beta; \pi)$	Thm 2.2
largest pole of L_p	critical inverse temperature $\beta_c(\pi_p)$	Thm 2.2
temperedness (Ramanujan) at p	$\beta_c(\pi_p) = w/2$	Thm 2.2
Eisenstein locus	the only locus carrying KMS <i>states</i>	Thm 2.5
Rankin–Selberg $L(s, \pi \times \tilde{\pi}')$	Kakutani/Hellinger obstruction Ξ_β	Thm 2.8
strong multiplicity one	mutual singularity of phases	Cor. 2.9
motivic weight w	width of the equilibrium/obstruction gap	Thm 2.11
Hecke recursion $TT_n = T_{n+1} + qT_{n-1}$	Ihara module $\mathbb{Z}[u^\pm]^n / (1 - uT_p + qu^2)$	Def. 3.3
Ihara zeta $Z_Y(u)$	Fitting ideal of the Ihara module	Prop. 3.4
Frobenius trace $a_p(f)$, $\#E_f(\mathbb{F}_p)$	invariant factors of $\text{SNF}(\Delta_p)$	Thm 3.5
Euler characteristic of Y (no arithmetic)	torsion of the boundary quotient	Thm 3.7

The reading of the table is the thesis of the paper. Along the top half — the thermodynamic rows — the correspondence is exact but weight-zero: it reproduces class field theory and Artin data, and Theorem 2.11 measures precisely how far it falls short above weight zero. Along the bottom half — the K -theoretic rows — the correspondence is insensitive to temperature and to positivity, and it does carry weight-two data, but only after the specialisation $u = 1$ that separates the Laplacian channel from the boundary channel. An algorithmic non-commutative class field theory for GL_2 is therefore not one bridge but two, and Bass’s identity (3.2) is the only thing that connects them.

ACKNOWLEDGEMENT OF PROVENANCE AND CAVEATS

This is an unrefereed preprint and no priority check against the literature has been carried out. The following are used as known and are not proved here: the Satake isomorphism and the Hecke recursion; Ihara’s theorem and Bass’s determinant identity; Kirchhoff’s matrix-tree theorem and the theory of the critical group; the Eichler–Shimura and Jacquet–Langlands correspondences; Kakutani’s dichotomy for product measures; Jacquet–Shalika non-vanishing for Rankin–Selberg L -functions; Renault’s theory of groupoid C^* -algebras and the correspondence between KMS states and conformal measures; Cuntz–Krieger and Cuntz–Pimsner K -theory and Pimsner’s exact sequence. The computations of Table 1, and the spectral and torsion-density computations of Remark 3.14, are exact or numerically controlled and were carried out independently of the statements they confirm. Theorem 3.12 rests on Lyons’ spanning-tree asymptotics [1], which we quote and do not prove, and Remark 3.14 on the Kesten–McKay spectral distribution [2, 3]. The one substantive gap left open is the one isolated in Remark 3.6: the Laplacian channel is a module and an index, not yet a C^* -algebra.

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